

ANJALI AGGARWAL

<mailto:anjali12agg@gmail.com> · (917) 456-2142 · [LinkedIn](#) · [GitHub](#) · [Portfolio](#)

EDUCATION

Texas Tech University

Honors College | Dean's List

B.S. in Computer Science & Mathematics Minor

May 2027

GPA: 3.6

Relevant Coursework: Data Structures & Algorithms, Computer Architecture, Operating Systems, Software Engineering, Database Systems, Discrete Computational Structures

TECHNICAL SKILLS

Developer Tools & Frameworks: Python, Java, C++, JavaScript, SQL, React, Node.js, Flask, Storybook, AWS, Git, Docker, Linux, PyTorch, TensorFlow, JSON Schema
AI/ML & Product: Machine Learning, Deep Learning, NLP, Computer Vision, LLM Pipelines, Prompt Engineering, Design Systems, WCAG 2.1 AA, Figma, Agile/Scrum
Certifications: AWS Cloud Practitioner, Anthropic AI, Google Cloud, Microsoft AI & ML, IBM Agile & Scrum, NYU Cybersecurity

PROFESSIONAL EXPERIENCE

Q2 Software

May 2026 – August 2026

AI Technical Product Management Intern | Austin, TX

- Documented 41 UI components for the Helix design system, covering tokens, variants, states, and WCAG 2.1 AA standards.
- Built an AI-assisted automation pipeline to generate, test, and validate component docs at scale, reducing documentation time by ~90%.
- Collaborated cross-functionally with design, engineering, and product teams to establish a single source of truth across Helix products.

AI Design Systems Software Engineer Intern | (Part-Time)

August 2026 – May 2027

- Coding out all 41 Helix components in React to be production-ready and deployment-ready, migrating docs from Confluence to Helix Storybook.
- Designing a structured schema per component and building automation to keep manifests in sync as documentation evolves.
- Standardizing the process for updating and retiring components, closing accessibility and token gaps, and drafting AGENTS.md instructions to scale the system.

Amazon — Operational Strategy & People Analytics

June 2025 – August 2025

Extern

- Developed Python workflows using NLP to collect, clean, and analyze employee voice data, identifying root causes of turnover.
- Collaborated with cross-functional teams to design a pilot-ready, data-driven retention intervention.
- Presented strategic insights directly to Amazon stakeholders, strengthening consulting and executive communication skills.

Cyber-Aerial Computing Lab — Texas Tech University

August 2026 – Present

Undergraduate Researcher

- Conducting hands-on research in drone-assisted computing using the Crazyflie 2.1+ platform, completing required safety certification for lab experiments.
- Designing and testing drone-based solutions across aerial networks, path planning, and cybersecurity countermeasures.
- Completing weekly hands-on labs with the Crazyflie 2.1+, progressing from flight setup to network and sensor-based navigation milestones.

Nanomagnetic Biosensors & Devices Group (Wu Research Group) — Texas Tech University

April 2025 – August 2026

Undergraduate Researcher

- Developed a deep learning model in Python to classify 12-lead ECG signals with ~85% accuracy across six heartbeat categories.
- Analyzed 10,000+ ECG samples from open-source medical datasets, improving signal clarity by 30% through custom filtering pipelines.
- Collaborated with the research team to evaluate clinical relevance of model predictions for cardiovascular diagnosis.

VZA Company Inc.

May 2025 – July 2025

Software Developer Engineer Intern

- Architected and deployed a Flask-based web application to automate shipment tracking, reducing manual data entry by 40%.
- Implemented real-time data validation to improve accuracy and streamline logistics workflows end-to-end.
- Built backend logic using SQL and JavaScript to manage inventory updates and generate summary reports for operational insights.

PROJECTS

Automatic ECG Diagnosis — Wu Research Group

April 2025- July 2026

- Developed an AI-assisted diagnostic system using a 1D ResNet architecture to classify six heart conditions from 12-lead ECG signals.
- Built a NumPy-based synthetic ECG generator producing 1,200+ labeled samples to enable training without clinical data.
- Designed a modular PyTorch pipeline with configurable training, evaluation, and ONNX export for deployment flexibility.

AI Chatbot System — HackWesTX

September 2024

- Built an NLP pipeline using TF-IDF and a linear classifier for intent recognition, achieving 90%+ accuracy on a curated validation set.
- Implemented regex-based entity extraction to enable dynamic, context-aware responses across real-time user queries.
- Served a responsive HTML/CSS/JavaScript chat interface through Flask to simulate a virtual help-desk experience.

Shipment Tracking Automation Tool — VZA Company Inc.

May 2025 – July 2025

- Built a Flask and SQLAlchemy application with real-time input validation and a background scheduler for asynchronous status updates.
- Designed a normalized SQL schema across shipments and events tables to support high-volume tracking and carrier event logging.
- Implemented automated CSV report generation with dynamic filters using Pandas to deliver tailored insights for operations managers.

AWARDS & RECOGNITION

HackWesTX — West Texas's Biggest Hackathon

May 2024 – Present

Sponsorship Lead / Competitor

- Led the largest edition of HackWesTX, drawing 300+ applications and raising over \$15,000 in sponsorships from companies including Patterson-UTI and Chevron.
- Competed as a developer, collaborating with a team to build and present a tech project under time constraints.

FIRST Tech Challenge (FTC) Robotics

January 2025 – May 2025

Programmer, Outreach Lead

- Programmed autonomous robot systems using computer vision pipelines (Easy OpenCV, April Tags).
- Won the Division Inspire Award at the FTC World Championship, the highest honor for engineering excellence and team impact.